

Question 1 (1 point)**Answer: D.** S_1 is false: take $L_1 = L_2 = \{a^n b^n\}$, both CFL. S_2 is false: take L_2 any CFL (e.g., $\{a^n b^n\}$) and L_1 a regular subset (e.g., a^*), then L_1 is CFL. S_3 is false: take $L_1 = \text{(CFL)}$ and $L_2 = \{a^n b^n c^n\}$ (not CFL) with $L_1 \subseteq L_2$.None of S_1, S_2, S_3 hold in general. Counterexamples: e.g. CFL $\{a^n b^n\}$ shows failures. Therefore, none are always true.**Question 2** (1 point)**Answer: C.**(A) False: Σ^* is regular but has non-regular subsets (like $\{a^n b^n\}$).

(B) False in general.

(C) True: any infinite regular language has an infinite regular subset.

(D) False: arbitrary subsets of a CFL need not be CFL.

Question 3 (1 point)**Answer: A.**Option A : $L_1 \cap L'_1 = \emptyset$, which is a DCFL.Option B : $L_2 \cap L_3$ is not necessarily DCFL.Option C : $L_3 \cup L_4$ may not be DCFL (union not closed).Option D : $L_3 \cap L_4 = \{a^n b^n a^n\}$, which is not context-free.**Question 4** (1 point)**Answer: B.**

A DTM has at most one next move from any configuration, so it cannot branch to multiple configurations; B is false.

Other listed options are consistent with DTM properties. Option A is true (DTM and NDTM are equivalent in computability). Option C is also true (multiple accept states allowed). Option D can be true when the DTM is a decider.

Question 5 (1 point)**Answer: D.**

Options A and B are standard Turing Machine capabilities. Option C is true: a given Turing Machine has a fixed finite set of states independent of input, but the number of configurations depends on input/tape contents. Thus, D is false, while other options are valid.

Question 6 (1 point)**Answer: A.** $L_1 = \{a^n b^m a^{n+m}\}$ is CFL (can be handled by PDA, that can push for the first a-block, push again for each b, then pop on the final a-block). $L_2 = \{0^{2n} 1^n\}$ is CFL (grammar $S \rightarrow 00S1 \mid \epsilon$).

Hence both are CFL.

Question 7 (1 point)**Answer: B.**

DTMs and NDTMs are equivalent in language-recognition power (both recognize exactly the RE languages and decide the recursive ones). So options A and C are false, and option D misstates the classes for DTMs.

Question 8 (1 point)**Answer: D.**Union: CFLs are closed under union, so $L_1 \cup L_2$ is CFL.Intersection: CFLs are not closed under intersection, even with DCFL, so $L_1 \cap L_2$ may not be CFL.

Hence option D is correct.